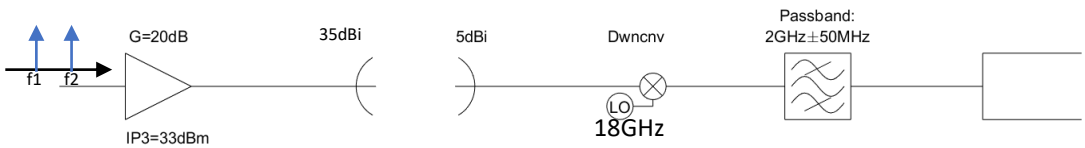
POLITECNICO MILANO 1863 Academic year 2023/2024	Course RF SYSTEMS	**Student ID or Persona code**
	Signature	**Student Name, Surname**
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Exercise 2 (6 points)

A power amplifier in a transmitter provides a gain of 20dB and has an output IP3 of 33dBm (pure 3rd order non linarity). The power amplifier is fed by a two-tone signal centered at 22GHz with spacing of 4 GHz between the two tones. Each tone has average power 5dBm.

a) What is the spectrum at the output of the PA, between 15GHz and 30GHz, specifying frequencies and amplitudes in dBm (neglecting compression of the main tones)?

b) The PA is connected to a 35dBi-gain antenna and acts as a jammer to interfere with a victim receiver. The victim receiver operates at 20GHz at a distance of 1500m, with a 5dBi-gain receiving antenna, perfectly reciprocally aligned toward the jammer. All components are matched at 50Ω



What is the received power of the 20GHz tone at the antenna port of the victim?
The receiver uses a local oscillator at 18GHz to internally downconvert the received signals to an IF of 2GHz with bandwidth 100MHz. Observe that the receiver has no image filter and assume it does not introduce any further intermodulation. Besides the 20GHz one, will other tones among the transmitted ones be downconverted to the same IF (indicate only the RF frequencies of such interfering tones, not their power)?

SOLUTION

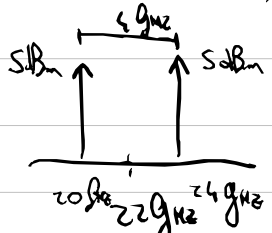
The signal at the input of the amplifier is 20GHz (5dBm) and 24GHz (5dBm).
At the output we also have the third-order intermodulation products: 2f1-f2=16GHz, f1=20GHz, f2=24GHz, 2f2-f1=28GHz
The main tones (20 and 24GHz) will only be amplified, neglecting compression: $P_{w1}=P_{w2}=25$ dBm
The intermod. products follow $P_{2w1-w2}=P_{2w2-w1}=3P_{w1}-2IP3 = 9$dBm

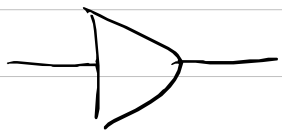
The 20GHz tone at 25dBm will be transmitted according to the link budget equation:
PRx=PTx+GTx-FSL+GRx, with FSL=20Log(4πd/λ)=122dB
Thus the tone at 20GHz will be received as PRx=25dBm+35dBi-122dB+5dBi=-57dBm.

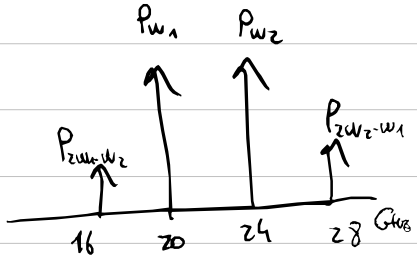
The receiver will receive at its antenna all 4 tones and downconvert them:
16GHz→downconverted to 2GHz: falls into the IF of the victim
20GHz→downconverted to 2GHz: falls into the IF of the victim
24GHz→downconverted to 6GHz: does not fall into IF of the victim
28GHz→downconverted to 10GHz: does not fall into IF of the victim

EX 2 — h 12:18 → 11 MIN

$G = 20\,dB$, $IP_3 = 33\,dBm$







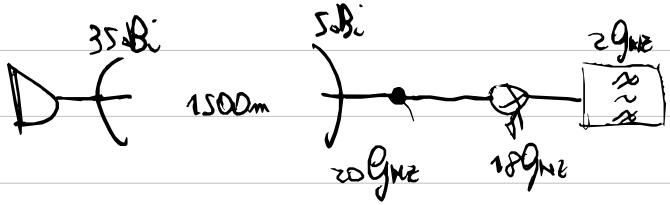
a)

$$P_{2\omega_2-\omega_1} = P_{2\omega_1-\omega_2} = 3P_{\omega_i} - 2IP_3$$

$$P_{\omega_2} = P_{\omega_1} \simeq G + P_{\omega_{1IN}} = 25\,dBm$$

$$\Rightarrow P_{2\omega_2-\omega_1} = (75-66)\,dBm = 9\,dBm = P_{2\omega_1-\omega_2}$$

b)



$$P_{\omega_1\,ANT} = P_{TX} + G_{TX} - FSL + G_{RX} \qquad FSL = 20\log\frac{4\pi d}{\lambda} = 122\,dB$$

$$= 25\,dBm + 35\,dBi - 122\,dB + 5\,dBi = -57\,dBm$$

$16\,GHz \rightarrow 2\,GHz,\ 34\,GHz$

$20\,GHz \rightarrow 2\,GHz,\ 38\,GHz$

$24\,GHz \rightarrow 6\,GHz,\ 42\,GHz$

$28\,GHz \rightarrow 10\,GHz,\ 46\,GHz$

The 16 GHz tone will also be downconverted to the same 2 GHz IF as the 20 GHz tone

h 12:29